

Git HandsOn - 4

Aim:

To understand Git merge conflict resolution by creating conflicting changes in two branches and resolving them manually.

Objectives:

- Create a new branch.
- Make different changes to the same file in two branches.
- Merge branches and resolve conflicts.
- Update the .gitignore file.
- Delete the merged branch.
- View the commit history.

Commands Used:

```
git branch GitWork
git checkout GitWork
git add hello.xml
git commit -m "Added hello.xml in GitWork"
git checkout main
git add hello.xml
git commit -m "Added hello.xml in main"
git merge GitWork
git add hello.xml
git commit -m "Resolved merge conflict"
git add .gitignore
git commit -m "Added backup files to gitignore"
git branch -d GitWork
git log --oneline --graph --decorate
git push origin main
```

Outputs:

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
* master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch GitNewBranch

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
  GitNewBranch
* master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch -a
  GitNewBranch
* master
remotes/origin/master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git branch
* GitNewBranch
  master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ echo This is my new branch > branch.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ ls
branch.txt  logs/  sample.log  welcome.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch e3c17da] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git diff master GitNewBranch
```

MINGW64:/c/Users/HP/Desktop/GitDemo

```
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..388da8a
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This is my new branch
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git merge GitNewBranch
Updating 2bf59b3..e3c17da
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --online --graph --decorate
fatal: unrecognized argument: --online
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was e3c17da).
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
* master
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 355 bytes | 71.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GayathriVanga/GitDemo.git
 2bf59b3..e3c17da master -> master
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.
```

nothing to commit, working tree clean

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch GitWork
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git checkout GitWork
Switched to branch 'GitWork'
```

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ git branch
* GitWork
```

```
MINGW64:/c/Users/HP/Desktop/GitDemo
master
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ echo "<message>Hello from GitWork Branch</message>" > hello.xml
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ git commit -m "Added hello.xml in GitWork"
[GitWork 8774fd7] Added hello.xml in GitWork
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ echo "<message>Hello from Master Branch</message>" > hello.xml
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git commit -m "Added hello.xml in master"
[master 264fddf] Added hello.xml in master
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --online --graph --decorate --all
fatal: unrecognized argument: --online
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git diff master GitWork
diff --git a/hello.xml b/hello.xml
index e3bbec4..234c289 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
-<message>Hello from Master Branch</message>
+<message>Hello from GitWork Branch</message>
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
```

```
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master|MERGING)
$ notepad hello.xml

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master|MERGING)
$ git add hello.xml

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master|MERGING)
$ git commit -m "Resolved merge conflict in hello.xml"
[master b26acf8] Resolved merge conflict in hello.xml

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ notepad .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git add .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git commit -m "Added backup files to gitignore"
[master a8ac4f6] Added backup files to gitignore
 1 file changed, 1 insertion(+)

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
  GitWork
* master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was 8774fd7).

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --online --graph --decorate
fatal: unrecognized argument: --online

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --oneline --graph --decorate
* a8ac4f6 (HEAD -> master) Added backup files to gitignore
* b26acf8 Resolved merge conflict in hello.xml
| \
| * 8774fd7 Added hello.xml in GitWork
* | 264fddf Added hello.xml in master
| /
* e3c17da (origin/master) Added branch.txt in GitNewBranch
* 2bf59b3 Added gitignore file
```

MINGW64:/c/Users/HP/Desktop/GitDemo

```
$ git status
On branch master
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ notepad .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git add .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git commit -m "Added backup files to gitignore"
[master a8ac4f6] Added backup files to gitignore
 1 file changed, 1 insertion(+)

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
  GitWork
* master

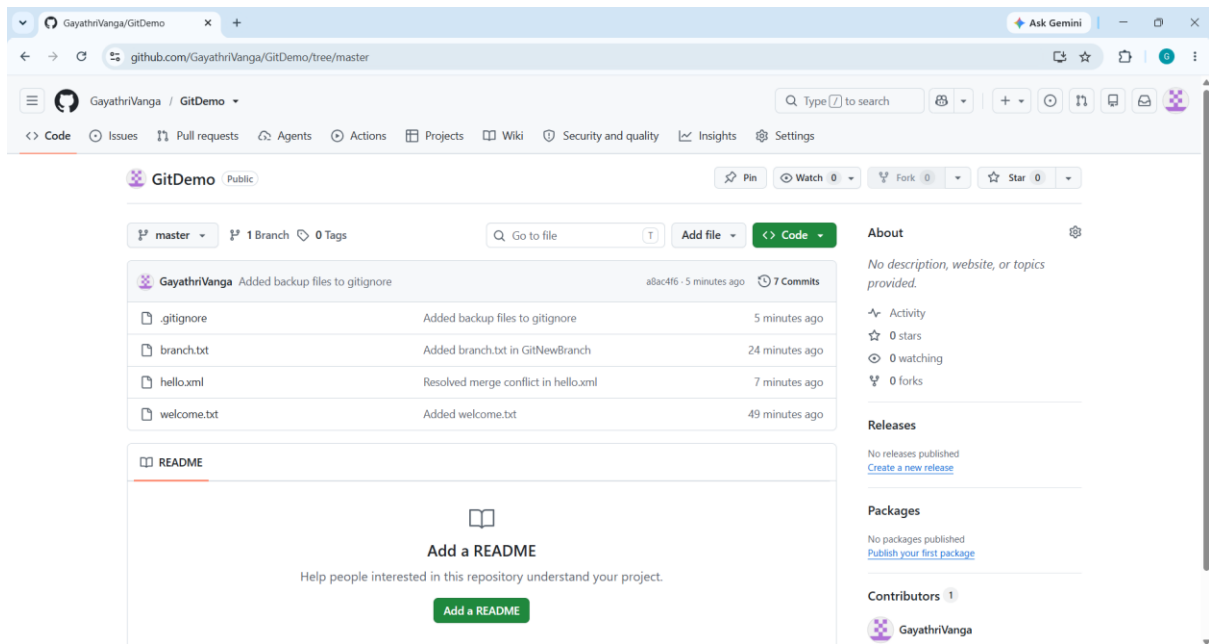
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was 8774fd7).

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --online --graph --decorate
fatal: unrecognized argument: --online

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --oneline --graph --decorate
* a8ac4f6 (HEAD -> master) Added backup files to gitignore
* b26acf8 Resolved merge conflict in hello.xml
| \
| * 8774fd7 Added hello.xml in GitWork
* | 264fddf Added hello.xml in master
| /
* e3c17da (origin/master) Added branch.txt in GitNewBranch
* 2bf59b3 Added gitignore file
* 5f4fe8f Added welcome.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git push origin master
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (12/12), 1.19 KiB | 134.00 KiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
To https://github.com/GayathriVanga/GitDemo.git
   e3c17da..a8ac4f6  master -> master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$
```



Result:

Successfully created a merge conflict by modifying the same file in two branches, resolved the conflict manually, committed the merged changes, updated the .gitignore file to ignore backup files, deleted the merged branch, and pushed the final changes to the GitHub repository.